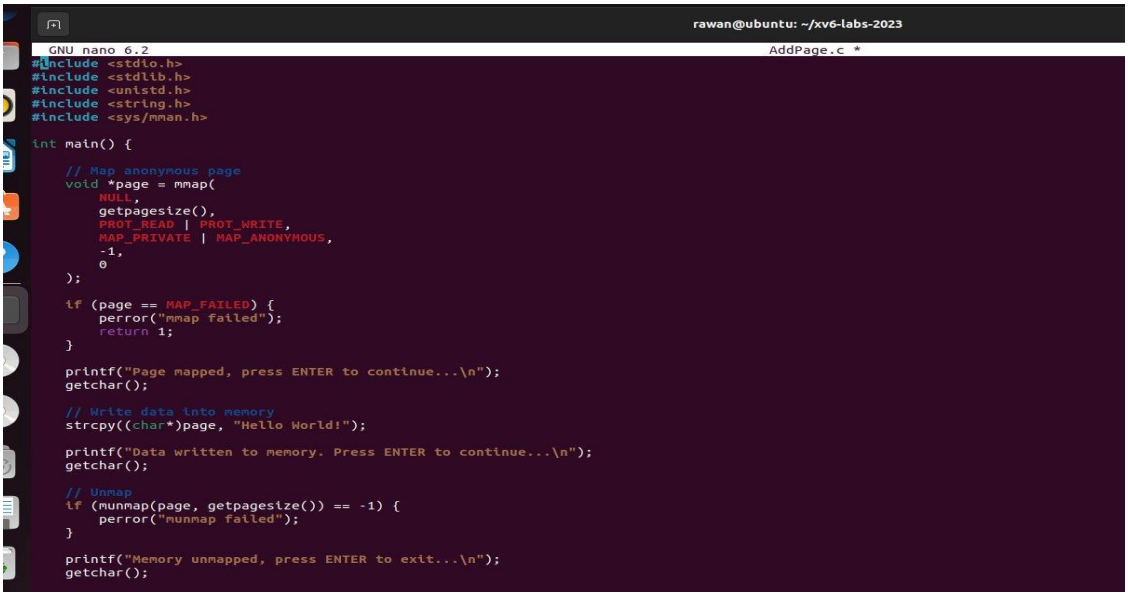
Implementation

Task 1 – Mapping Memory using mmap()

This task demonstrates how a process can reserve a page in virtual memory using mmap().

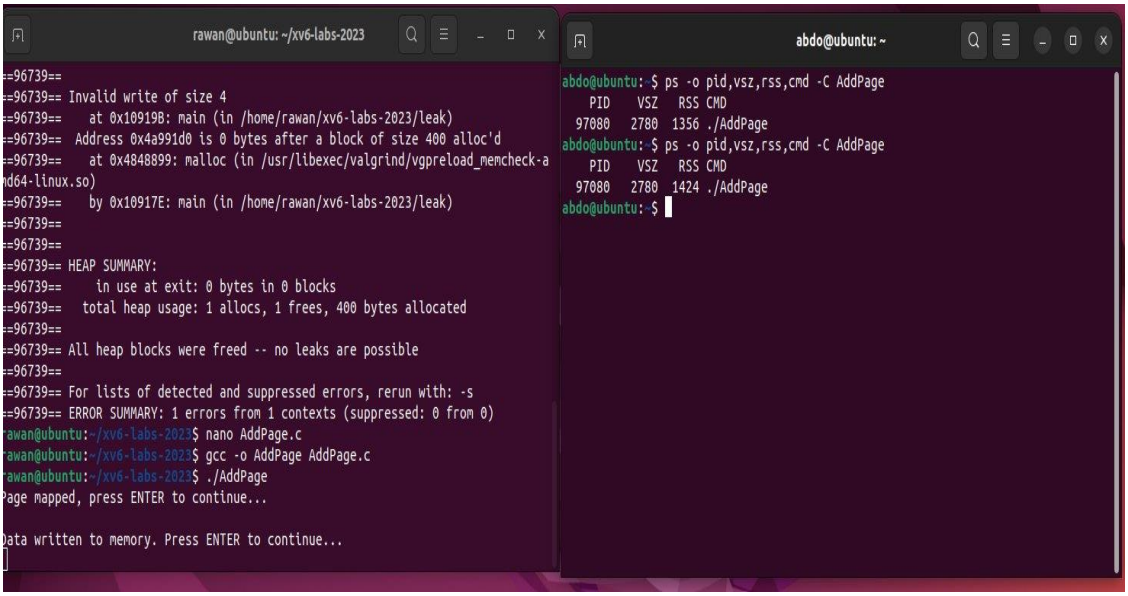


The program pauses before and after writing data to allow measuring memory usage.

Initially, the page is only reserved in virtual memory, not loaded into RAM.

Task 2 – Writing to the Mapped Page

After writing data into the mapped page, the system allocates real memory.



VSZ stays the same because no new virtual space is allocated.

RSS increases because the page is now physically stored in RAM after access.

Memory Behavior Explanation

Virtual memory is allocated when mmap() is called, but physical memory is only used when accessed.

This is known as demand paging, where memory is used only when needed.

Once the page is accessed, the OS loads it into RAM, increasing RSS.

Summary

Before mapping: minimal memory usage.

After mapping: virtual memory increases.

After writing: physical memory increases.

After unmapping: memory returns to normal.